MIL-C-24643/15B
3 October 1986
SUPERSEDING
MIL-C-0024643/15A(SH)
8 November 1985
MIL-C-24643/15
28 September 1984

MILITARY SPECIFICATION SHEET

# CABLE, ELECTRICAL, 1000 VOLTS, TYPE LSDSGU (INCLUDING VARIATION LSDSGA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

Sizes 3 to 1/23 inclusive, extruded silicone rubber insulation

First - Copper conductor, uncoated (see table I for size).
Second - Extruded silicone rubber insulation (see table I for diameter).
Third - Glass braid.
Fourth - Braid covering (see table I for diameter). Standard identification code applied by method 1.

Sizes 1/23 to 400 inclusive, silicone rubber-glass tape insulation

First - Copper conductor, uncoated (see table I for size).
Second - Silicone rubber-glass tape insulation (see table I for diameter). Letter identification code applied by method 5.
Third - The two conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used to form a firm, well-rounded assembly.
Fourth - A binder.

1/ Size 23 may be glass tape or extruded silicone rubber insulation at the manufacturer's option.

(B) denotes changes.

AMSC N/A FSC 6145
DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited



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Construction variant type LSDGSU

Sizes 3 to 1/ 23 inclusive, extruded silicone rubber insulation

Fifth - The two conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used to form a firm, well-rounded assembly.
Sixth - Binder or combination binder and barrier.
Seventh - Cross-linked polyolefin jacket. (See table I for thickness.)

Sizes 1/ 23 to 400 inclusive silicone rubber-glass tape insulation

Fifth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking required.

Construction variant for type LSDSGA

(B) Fifth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking not required.
Sixth - Braided metal armor and paint.

1/ Size 23 may be glass tape or extruded silicone rubber insulation at the manufacturer's option.



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TABLE I. Details.

| 1/ Military part no. M24643/15 | 2/ Size | Conductor size | | Type's overall diameter (inches) | | | | Diameter over outer insulation tape nominal (inch) | Diameter over braid covering (max) (inch) | Diameter over insulation (min) (inch) | Jacket thickness nominal (inch) | Conductor resistance per 1000 feet at 25°C maximum (ohms) | Insulation resistance per 1000 feet (min) (megohms) | Cold bending mandrel diameter (max) (inches) | Accelerated service loading ampere |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | LSDSGU | | LSDSGA | | | | | | | | | |
| | | Navy standard | AWG | min | max | min | max | | | | | | | | |
| -01UN | 3 | | 16(class B) | | 0.391 | | 0.441 | ---- | 0.130 | 0.096 | 0.030 | 4.3 | 500 | ---- | ---- |
| -02UN | 4 | | 14(class B) | | 0.427 | | 0.477 | ---- | 0.143 | 0.112 | 0.030 | 2.68 | 500 | ---- | ---- |
| -03UN | 9 | | 10(class B) | | 0.544 | | 0.594 | ---- | 0.187 | 0.154 | 0.040 | 1.08 | 500 | ---- | ---- |
| -04UN | 14 | | 9(class B) | | 0.670 | | 0.720 | ---- | 0.262 | 0.230 | 0.040 | 0.859 | 500 | ---- | 100 |
| -05UN | 23 | | 7(class B) | | 0.781 | | 0.831 | 0.274 | 0.310 | 0.271 | 0.050 | 0.543 | 500 | ---- | 130 |
| -06UN | 50 | | 3(class C) | | 0.911 | | 0.961 | 0.334 | ---- | ---- | 0.050 | 0.21 | 200 | 12 | 215 |
| -07UN | 75 | | 1(class C) | | 1.074 | | 1.124 | 0.407 | ---- | ---- | 0.050 | 0.134 | 175 | 13 | 285 |
| -08UN | 100 | | 0(class D) | | 1.167 | | 1.217 | 0.453 | ---- | ---- | 0.050 | 0.106 | 160 | 15 | 335 |
| -09UN | 200 | | 0000 (class D) | | 1.583 | | 1.633 | 0.634 | ---- | ---- | 0.060 | 0.053 | 125 | 20 | 525 |
| -10UN | 300 | | 300 MCM (class D) | | 1.841 | | 1.891 | 0.748 | ---- | ---- | 0.075 | 0.0377 | 110 | 23 | 700 |
| -11UN | 400 | 400(127) | | | 2.069 | | 2.119 | 0.862 | ---- | ---- | 0.075 | 0.0273 | 100 | 25 | 835 |

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSDSGU and LSDSGA.



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EXAMINATION AND TESTS:

| | Requirements |
|---|---|
| **Basic electrical:** | |
| Conductor resistance - ohms/1000 feet at 25°C, maximum | (see table I) |
| Voltage withstand - volts, root mean square, minimum | |
| Sizes 3 through 9 | 3000 |
| Sizes 14 and larger | 5000 |
| Insulation resistance - megohms/1000 feet, minimum | (see table I) |
| Conductor continuity | No failure |
| **Group A:** | |
| Visual and dimensional | No failure |
| Watertightness - see MIL-C-24643 for limits of water leakage | No failure |
| Abrasion resistance - sizes 3 through 23 (extruded silicone rubber) scrapes, minimum | 250 |
| (B) Crack resistance - sizes 3 through 23 (extruded silicone rubber) | No damage |
| **Group B:** | |
| Gas flame - 1 hour | No failure |
| Drip - 95 ± 1°C | Zero |
| Tear - pounds per inch thickness, minimum (ASTM D 470) | 35 |
| Cold bending, cable, sizes 50 and larger - (see table I for mandrel diameter) | No damage |
| Physicals (unaged) | |
| Insulation (extruded) | |
| Tensile strength - $lbf/in^2$, minimum | 750 |
| Elongation - percent, minimum | 125 |
| Jacket (cable) | |
| Tensile strength - $lbf/in^2$, minimum | 1300 |
| Elongation - percent, minimum | 160 |
| **Group C:** | |
| Physicals (aged) | |
| Jacket (cable) | |
| Air oven | |
| Tensile strength - percent of unaged, minimum | 60 |
| Elongation - percent of unaged, minimum | 60 |
| Hot oil immersion | |
| Tensile strength - $lb/in^2$, minimum | 50 |
| Elongation - percent, minimum | 50 |
| Shrinkage | No failure |
| Heat distortion - percent of unaged, maximum | 30 |
| Permanence of printing (conductor) | |
| Sizes 3 through 23 (extruded silicone rubber) - cycles, minimum | 50 |
| Permanence of printing (jacket) - cycles, minimum | 250 |
| Cable filler removability | No failure |
| Armor (type LSDSGA only) - conformance to material, construction and coverage | No failure |



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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

| | Requirements |
|---|---|
| Cold working (minus 20 ± 2°C) | No damage |
| Gas flame - 3 hours | No failure |
| Specific gravity of extruded silicone insulation, maximum | 1.55 |
| Accelerated service - sizes 14 and larger (see table I for load current) | No failure |
| Aging and compatibility (cable)(125 ± 5°C) | No failure |
| (B) Abrasion resistance (jacket) - scrapes, minimum | 75 |
| Acid gas equivalent - percent, maximum | |
| Jacket | 2 |
| Fillers | 2 |
| Insulation | 18 |
| Halogen content - percent, maximum | |
| Jacket | 0.2 |
| Fillers | 0.2 |
| Flame propagation (cable) | No failure |
| Immersion (jacket) | |
| Tensile strength - percent of unaged, minimum | 50 |
| Elongation - percent of unaged, minimum | 50 |
| Smoke index, maximum | |
| Jacket | 25 |
| Fillers | -- |
| (B) Insulation | 35 |
| Toxicity index, maximum | |
| Jacket | 5 |
| Fillers | 5 |
| Insulation | 1.5 |
| Durometer (jacket) - (type A) hardness, minimum | 80 |
| Weathering (jacket) | No failure |

UNIT ORDERING LENGTHS:

| Type and size | Feet (nominal) |
|---|---|
| LSDSGU-3 | 2500 |
| LSDSGU-4 through 23 | 2000 |
| LSDSGU-50 | 1500 |
| LSDSGU-75 through 100 | 1000 |
| LSDSGU-200 through 400 | 800 |

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NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-15)

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